

6(a)	<u>sum of</u> current(s) into junction = <u>sum of</u> current(s) out of junction or (algebraic) sum of current(s) at a junction is zero	B1
6(b)(i)	$I = 3.6 - 2.1$ $= 1.5$	C1
	$V = 4.4$	C1
	$R = 4.4 / 1.5$ $= 2.9 \Omega$	A1
6(b)(ii)	$12.0 = 4.4 + 3.6r$ or $12.0 = 3.6 (1.2 + r)$	C1
	$r = 2.1 \Omega$	A1
6(b)(iii)	$t = (470 \times 10^3 - 240 \times 10^3) / (12 \times 3.6)$	C1
	$= 5300 \text{ s}$	A1
6(b)(iv)	$I = Anvq$ ratio = $(360A / A) \times (2.5n / n)$ or 360×2.5	C1
	$= 900$	A1